PERFORMANCE EVALUATION FOR SPOKEN LANGUAGE OF A SYNTACTIC ANALYSIS METHOD USING INDUCTIVE LEARNING

YOSHIYUKI MASATOMI, KENJI ARAKI, and KOJI TOCHINAI

Graduate School of Engineering, Hokkaido University, Sapporo, Hokkaido, Japan 0608628

Abstract

In this paper, we propose a syntactic analysis method using inductive learning from examples. In our proposed method, the system acquires the parsing rules using the examples of parsing result. And the system parses Japanese sentences using the acquired parsing rules. We consider that our proposed method can resolve problems of the Rule-based approach and the Example-based approach. Moreover, we consider that this method can be applied to the other languages. We performed the experiment using transcripts of simulated spoken dialogue corpus. We consider the result of experiment about the ability of our proposed method.

Keywords

Inductive learning, Segmentation, Multi-language, Syntactic analysis.

1 Introduction

In natural language processing, parsing is very important in analyzing sentences. Therefore, a great deal of research concerning the syntactic analysis method has been focused on. The main method of parsing is the Rule-based method [1, 2]. However, the Rule-based method cannot deal adequately with various linguistic phenomena due to its use of limited rules. Therefore, its parsing ability is still low.

To resolve this problem, Example-based approach has become a common technique for natural language processing, especially in machine translations [3, 4, 5]. Example-based parser was recently proposed [6]. The correct parsing rate and the accuracy of Example-based parser is high because this method adapts to data automatically. However, this method requires many examples of parsed sentences in order to parse sentences correctly.

Moreover, the syntactic analysis methods which utilize decision tree were proposed [7, 8, 9]. These methods parse sentences stochastically by using a tree structure. In addition to the methods mentioned above, a method to extract grammar from a corpus automatically has been proposed [10]. However, the ability of the parser based on statistical method depends on sample sentences.

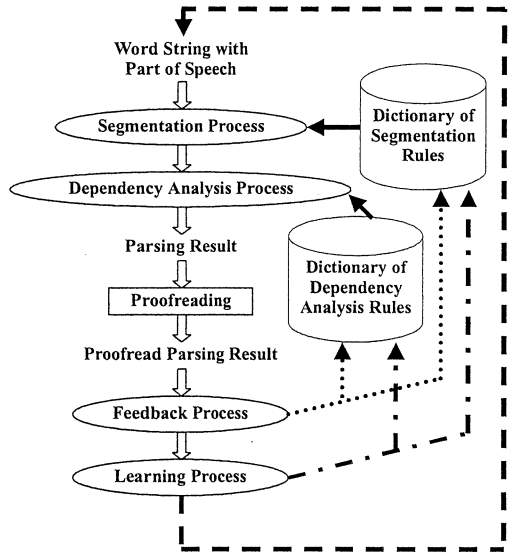


Figure 1 Process

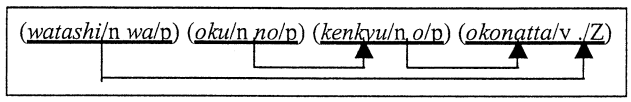
In order to resolve the problems associated with these syntactic analysis methods, we proposed a syntactic analysis method using the inductive learning. In this method, parsing rules are inductively acquired and parsing results are produced through the use of these rules. This method does not decide the solution stochastically. We consider it is possible to acquire parsing rules automatically, and extract parsing rules of various abstraction degrees from the acquired rules recursively. In our proposed method, parsing rules are extracted from word strings by using the inductive learning. Moreover, we consider that the parser based on our proposed method can improve accuracy by applying parsing rules of the most suitable abstraction degree to the input word string. A parser based on this method will continuously evolve to higher level of quality, and will actively adapt to the input data. Because the learning algorithm does not depend on the specific language in this method, we consider it can be applied to various languages.

0-7803-7087-2/01/$10.00 © 2001 IEEE

Table 1 An example word string with part of speech and segments.

Japanese	*watashiwaokunokenkyuwookonatta.*								
English	I did many kinds of research.								
Word string with POS	*watashi*/n	*wa*/p	*oku*/n	*no*/p	*kenkyu*/n	*wo*/p	*okonatta*/v	. /Z	
Segments	*watashi*/n *wa*/p		*oku*/n *no*/p		*kenkyu*/n *wo*/p		*okonatta*/v . /Z		

n:Noun, p:Particle, v:Verb, Z:Period



( ):Segment, n:Noun, p:Particle, v:Verb, Z:Period

Figure 2 An example of parsing result.

Table 2 An example of segmentation rules.

No variable	(*oku*/n *no*/p) (*kenkyu*/n *o*/p)
Not all valiable	(@0/n *wa*/p) (@1/v ./Z)
All variable	(@2/n @3/p) (@4/v @5/Z)

( ):Segment, n:Noun, p:Particle, v:Verb, Z:Period

2 Outline

We proposed a syntactic analysis method using the inductive learning. Figure 1 shows the outline of this method. The system based on our proposed method parses an input sentence as mentioned below.

First, a user inputs a word string with a part of speech [1]. This word string is obtained by recognizing the word by using the inductive learning [11] and tagged with part-of-speech (POS). Table 1 shows an example of a Japanese sentence, its word string with POS, and its segments[2]. Second, the system segments the word string using segmentation rules. Third, the system decides a dependency relation between segmentations using dependency analysis rules. These segmentation and dependency analysis rules are called parsing rules in this paper and acquired in a learning process. Figure 2 shows an example of the representation of the parsing result from this parser. Fourth, the user proofreads the parsing result produced in the aforementioned processes. Fifth, the system determines the fitness degree and performs the selection process of parsing rules. Lastly, in the learning process, various parsing rules are extracted from the input word string and its proofreading parsing result by the inductive learning. Moreover, new parsing rules are recursively extracted from the acquired parsing rules in this process. By repetition of these processes, a dictionary of parsing rules becomes accurate and sufficient for parsing. Thus, the system continuously improves its quality of parsing. Moreover, we consider it can be applied to various languages, because the learning algorithm does not depend on the specific language in this method.

2.1 Segmentation process

In this process, the system segments the input word string using segmentation rules which acquired in the learning process. Table 2 shows an example of segmentation rules.

The system adapts the segmentation rules as follows;

- The number of match words is maximum in the segmentation rules not included variables.
- The number of match words is maximum and the number of variables is minimum in the segmentation rules included variables.
- The number of match part-of-speech is maximum in the segmentation rules which all words are variables.

When plural segmentation rules can be applied, the rule, which has the maximum value in a credibility evaluation function, is applied. The credibility evaluation function (CEF) is as follows;

$$CEF = AF + \alpha \times CF - \beta \times EF \quad (1)$$

In this formula, AF is "Frequency of Appearance", CF is "Frequency of Correct Analysis", and EF is "Frequency of Erroneous Analysis". α and β are coefficients.

In addition, when there are plural rules which have the maximum value in the function, the system decides a rule as follows;

[1] In this paper, word strings are Japanese sentences.
[2] In this paper, Japanese are written in *italics*.

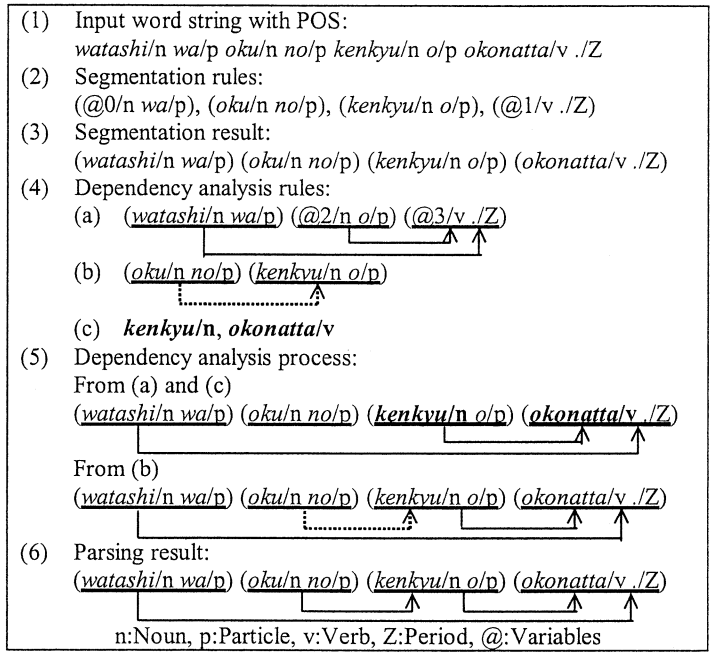
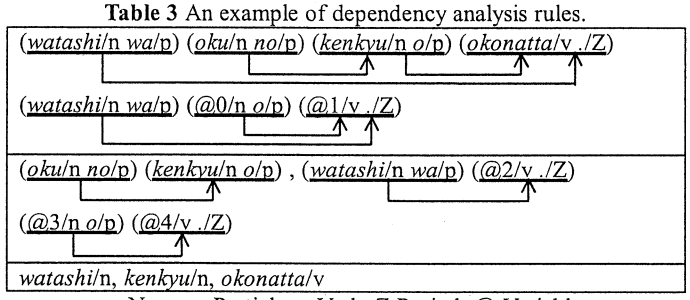


Figure 3 An example parsing process.

Table 3 An example of dependency analysis rules.



n:Noun, p:Particle, v:Verb, Z:Period, @:Variable

(1) The rule which has the minimum EF

(2) The rule which has the maximum CF

(3) The rule which has the maximum AF

(4) The rule which is the last in the dictionary

(1)~(3) in Figure 3 show an example of segmentation process.

2.2 Dependency analysis process

In this process, the system parses the segmentation results from segmentation process. Table 3 shows an example of dependency analysis rules. These rules are acquired in the learning process. The best match dependency analysis rule is sequentially applied to the input and a parsing result is produced. When plural dependency analysis rules can be applied, the rule, which has the maximum value in CEF, is applied.

In addition, when there are plural rules which have the maximum value in the function, the system decides a rule as follows;

(1) The match rule is the longest and maximum number of characters.

(2) The number of rules composed is minimum.

(3) The average of correct parsing rate of the parsing rules which compose is the highest one.

(4) The total of correct parsing degree of the parsing rules which compose is the highest one.

(5) The total of error-parsing degree of the parsing rules which compose is the lowest one.

(6) The dictionary procedure is last

(4)~(6) in Figure 3 shows an example of how the system produces a parsing result.

2.3 Feedback process

The procedure for the feedback process is as follows.

- The parser decides the common parts and the different parts between the parsing result and the proofreading parsing result.
- The common parts are assumed correct.
- The different parts are assumed incorrect.
- The parser adds 1 to the correct parsing frequency of the correctly analysed parsing rules.
- The parser adds 1 to the error-parsing frequency of the erroneously analysed parsing rules.

2.3 Learning process

In the learning process, the system uses the inductive learning and acquires the parsing rules from the word string with POS and the proofreading parsing result. The parsing rules are acquired from the common parts and the different parts between the word strings and the proofreading parsing results. The system recursively repeats this process until a new rule is not extracted from the acquired parsing rules [12].

In the case that a common degree (CD) is over a defined value, dependency analysis rules are acquired. The CD is as follows;

$$CD = \frac{CWN}{AWN} \times 100[\%]$$

In this formula, CWN is "Number of Common Words" and AWN is "Number of All Words". Common words are also same appearance order in the rules.

In addition, in the case that an abstract degree (AD) is under a defined value, new dependency analysis rules are extracted from abstracted rules. The AD is as follows;

$$AD = \frac{VN}{AWN} \times 100[\%]$$

In this formula, VN is "Number of Variables". Common words are also the same appearance order in the rules.

When the system acquired or extracted dependency analysis rules, the CD and the AD limit the number of rules. If there is no limit, the system acquired and extracted many rules which are not able to use in the dependency analysis process. Therefore, the system cannot process efficiently.

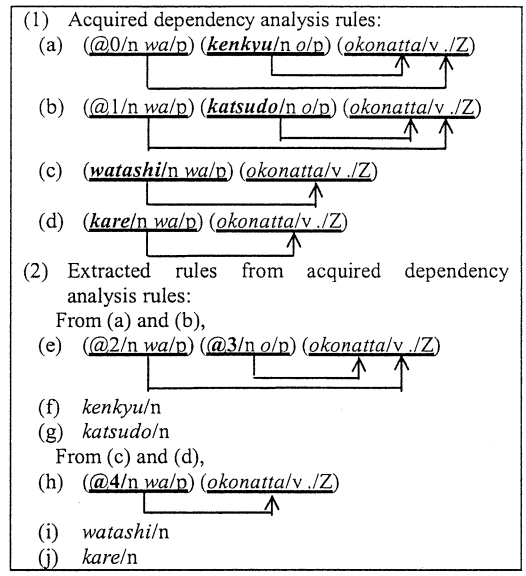


Figure 4 An example of inductive learning.

Figure 4 shows an example of the extracted dependency analysis rules from the acquired rules using the inductive learning. In this figure, there are the acquired dependency analysis rules (a)~(d) in the dictionary of rules. Between (a) and (b), the underlined parts are the different parts. The common parts do not require changing, and the system changes the different parts into variables. Thus, new rule (e) is extracted. Thereby, new rules (f) and (g) are extracted. In the same way, a new rule (h) is extracted from (c) and (d). Thereby, new rules (i) and (j) are extracted from the different parts.

3 Experiment

We performed the experiment using transcripts of simulated spoken dialogue corpus [13]. In this experiment, we used 1,748 sentences in the corpus. In these sentences, the total number of segments is 8,067, the average number of segments is 4.6 per a sentence, and the average number of words is 12.4 per a sentence. A content of corpus is about schedule. The content is same, however there are three kinds. Each corpus was prepared by different university which are Osaka University (OU), The University of Electro-Communications (UEC), and Kyoto University (KU). A single sentence is input to the system. That is, input sentences are unknown.

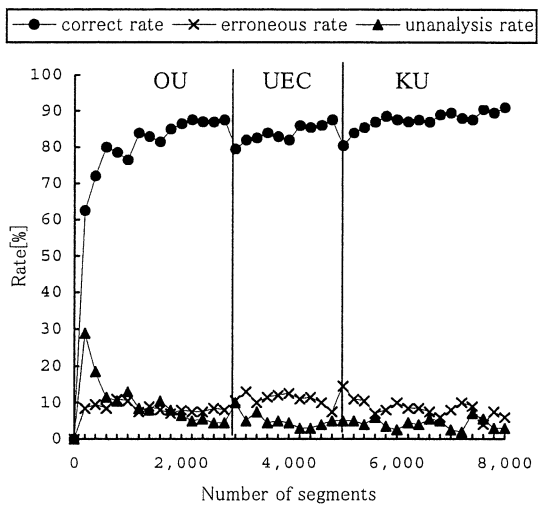


Figure 5 Change of segmentation rates.

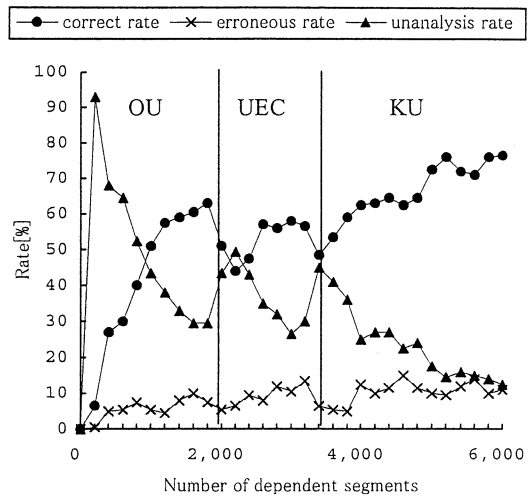


Figure 6 Change of dependency analysisrates.

Therefore, the system can dynamically adapt to the input sentences.

In this experiment, we concentrated only on the evaluation of the dependency relationship.

3.1 Procedure

The dictionaries of segmentation rules and dependency analysis rules are empty in the initial state. We experimented using the following procedures.

I. A Japanese sentence is changed into a word string with POS.

II. The word string with POS is inputted.

III. The system parsed the input with the dictionary of parsing rules and a parsing result is obtained[3].

IV. The feedback processing is carried out from the parsing result and the proofread parsing result.

V. The parsing rules are acquired by using inductive learning from pairs in the word string and the parsing result that has been proofread.

VI. New parsing rules are extracted from the parsing rules that have already been acquired using the inductive learning in step **V**.

VII. Step **VI** is recursively repeated until no new rules are extracted.

VIII. The above operations are repeated using 1,748 Japanese sentences.

In the step **V**, the number of sentences that the system learns is 100 sentences before the input sentence. Because, we consider that it is unnecessary to learn all sentences from the standpoint of human learning research.

We carried out preliminary experiments to decide the CD, the AD, α, and β. We defined each value as follows;

CD = 50[%], AD = 50[%], $\alpha = 1$, $\beta = 3$

3.2 Experiment results

Figure 5 shows the change of segmentation rates per 200 segments for this experiment. The last correct segmentation rate was about 90%. Moreover, Figure 6 shows the change of dependency analysis rates. The last correct dependency analysis rate between 5,801 and 6,000 dependency relationships was 76.5%. In these figures, two lines represent the change of corpus.

4 Discussion

In Figure 6, the accuracy of the dependency analysis results based on this method improves gradually. In the initial state, the dictionary of parsing rules is empty. Therefore, when the number of input sentences is low, the number of parsing rules is also low. Then, the parsing rules that can be applied to the input word string are not usually in the dictionary of parsing rules. Therefore, the dependency relationship cannot be determined during the dependency analysis process, and the correct dependency analysis rate is low when the number of input sentences is low. However, the number of parsing rules increases when the number of input sentences increases, and the number of parsing rules which can be applied to the input word string increases. As a result, the

[3] In this experiment, the result is the dependency relationship between segments.

dependency relationship can be determined during the dependency analysis process and the correct dependency analysis rate increases, as well. Therefore, our proposed method has the ability to improve accuracy of the dependency relationship gradually and increased effectiveness can be realized.

The correct segmentation rate and the correct dependency analysis rate temporary decreased when the input corpus changed. It is considered that the system was adapted to the corpus which had been input. Therefore, the system cannot analyse correctly until it is adapted to new input corpus. However, the correct dependency analysis rate gradually increases when the number of input sentences increases.

Here we consider the last correct dependency analysis rate between 5,801 and 6,000 dependency relationships. Its rate is 76.5%. This is not high rate compared with other methods. However, this result is the analysis rate for the spoken language which is difficult to analyse in other methods. Moreover, we compare the last dependency analysis rate with a baseline rate. We defined the correct dependency analysis rate in the case that all segments depended on next segment as the baseline rate. The baseline rate was 61.0% between 5,801 and 6,000 dependency relationships in experimental sentences. Therefore, we confirm that our proposed method has the ability in syntactic analysis for spoken language.

5 Conclusions

To resolve the problems which the Rule-based method, Example-based method and statistical method have, we proposed a syntactic analysis method which utilizes the inductive learning. In addition, we designed a system based on our proposed method. In the system, parsing rules were acquired by using the inductive learning, and recursively extracted from the acquired parsing rules. As a result, the parsing rules could be acquired automatically, and the system acquired the parsing rules of a variety of abstractions.

Moreover, we evaluated the performance of this system using spoken language corpus. In this experiment, the system applied the parsing rules of the most suitable abstraction degree to the input word strings. The last correct dependency analysis rate between 5,801 and 6,000 dependency relationships was 76.5%. This rate was 15.5% higher compared with the baseline rate. We confirmed that the correct dependency analysis rate increased gradually and our proposed method had the ability of syntactic analysis for spoken language.

To improve accuracy, we will examine the learning algorithm and applying algorithm of parsing rules. We also would like to apply this method to other languages as well.

References

1. E.Brill and P.Resnik, A Rule-based Approach to Prepositional Phrase Attachment Disambiguation, Proc. of the 15th COLING, 1994, 1198-1204.
2. U.Germann, A Deterministic Dependency Parser for Japanese, Asia-Pacific Association for Machine Translation, 1999, 547-555.
3. V.Sadler and Vendelmans, Pilot Implementation of a Bilingual Knowledge Bank, Proc. of the 13th COLING, 1990, 449-451.
4. C.Stanfill and D.Waltz, Toward Memory-based Reasoning, Communications of the ACM, 29(12), 1986, 1213-1228.
5. S.Sato and M.Nagao, Toward Memory-based Translation, Proc. of the 13th COLING, 1990, 247-252.
6. M.Al-Adhaileh, and T.E.Kong, A Flexible Example-based Parser Based on the SSTC, Proc. of the ACL, 1998, 687-693.
7. M.Haruno, S.Shirai, and Y.Ooyama, A Japanese Dependency Parser Based on a Decision Tree, Transactions of Information Proc. Society of Japan, 39(12), 1998, 3177-3186.
8. R.Bod, A Computational Model of Language Performance Data Oriented Parsing, Proc. of the 14th COLING, 1992, 855-859.
9. S.Mori and M.Nagao, A Stochastic Language Model Using Dependency, IEICE Technical Report, NL122-6, 1997.
10. K.Shirai, T.Tokunaga, and H.Tanaka, Automatic Extraction of Japanese Probabilistic Context Free Grammar from a Bracketed Corpus, Transactions of Natural Language Proc. Society of Japan, 4(1), 1997, 125-146.
11. K.Araki, and K.Tochinai, Acquisition Words by Inductive Learning and Recognition Words Using Certainty, IEICE Trans. J75-D-II(7), 1992, 1213-1221.
12. K.Araki, Y.Takahashi, Y.Momouchi, and K.Tochinai, Non-segmented Kana-kanji Translation Using Inductive Learning, IEICE Trans. J79-D-II(3), 1996, 391-402.
13. Grant-in-Aid for Scientific Research on Priority Areas (by Ministry of Education, Science, Culture, Japan), Research on Understanding and Generating Dialogue by Integrated Processing of Speech, Language and Concept, 1996.